上海财经大学 · 金融工程回忆版试卷

试题

一、Financial Derivatives

1. What are Financial Derivatives?
2. Silver current price ¥8,000/kg, 1 lot = 15 kg. Silver is a futures contract with leverage ratio 10. If our margin account has ¥300,000, how many lots can we short?
3. We short 5 lots at beginning. After a week, our margin account becomes ¥306,000. What is the current silver futures price?

二、Stochastic Calculus

1. What is the distribution of $W_1 + W_2 + W_4$?
2. Compute $\int_0^T W_t \, \mathrm{d}W_t$.
3. Assume

   $$X_t = 3t^2 + W_t^4 + ctW_t^2$$

   prove X_t is a diffusion process.
4. If the drift of X_t is always 0, what is the possible value of c?
5. Is geometric Brownian motion a diffusion process? Why?
6. How to convert the BSM to the risk-neutral measure?
7. What is the biggest difference between the Itô integral and ordinary calculus?

三、Risk-Neutral Pricing

1. What is the Black-Scholes model?
2. Consider an option where we pay ¥1 to its holder at maturity T, if and only if $S_T > K$. Write the payoff function of this option.
3. Use Black-Scholes option pricing to price this option.
4. If you hold a long American call option, what is your optimal exercise strategy?

答案与解析

一、Financial Derivatives

(1) What are Financial Derivatives?

Answer: A financial derivative is a contract whose value derives from an underlying asset (stock, bond, commodity, currency, rate, or index). They are used for hedging, speculation, and arbitrage.

Types:

- Forward: OTC agreement to buy/sell at future date at price fixed today
- Futures: Standardized, exchange-traded forward with daily mark-to-market
- Options: Right (not obligation) to buy (call) or sell (put) at strike K; European (at T) or American (any time)
- Swaps: Exchange cash flows (interest rate, currency, CDS)

Key characteristics: leverage, zero-sum (forwards/futures), nonlinear payoffs (options), mark-to-market (futures).

解析: Derivatives are called “derivative” because their price is derived from an underlying asset. They enable risk transfer between counterparties.

(2) Silver current price ¥8,000/kg, 1 lot = 15 kg. Silver is a futures contract with leverage ratio 10. If our margin account has ¥300,000, how many lots can we short?

Answer: 25 lots.

解析:

- Silver is a futures contract with leverage ratio 10, i.e. initial margin $= \frac{1}{10} = 10\%$
- Notional per lot: $15 \times 8000 = ¥120,000$
- Initial margin per lot: $0.10 \times ¥120,000 = ¥12,000$ per lot
- Number of lots: $¥300,\frac{000}{¥}12,000 = 25$ lots

(3) We short 5 lots. After a week, margin account becomes ¥306,000. What is the current silver futures price?

Answer: ¥7,920/kg.

解析:

- Account gain: $¥306,000 - ¥300,000 = ¥6,000$ (short position profits when price falls)
- Total quantity: $5 \times 15 = 75$ kg
- Price drop: $¥6,\frac{000}{75} = ¥80$ per kg
- New price: $¥8,000 - ¥80 = ¥7,920$ per kg

二、Stochastic Calculus

(1) What is the distribution of $W_1 + W_2 + W_4$?

Answer: $W_1 + W_2 + W_4 \sim N(0, 15)$.

解析: Rewrite as independent increments: $W_1 + W_2 + W_4 = 3W_1 + 2(W_2 - W_1) + (W_4 - W_2)$

Variance:

$$\begin{aligned} & 3^2 \cdot \mathrm{Var}(W_1) + 2^2 \cdot \mathrm{Var}(W_2 - W_1) + 1^2 \cdot \mathrm{Var}(W_4 - W_2) \\ = {} & 9 \cdot 1 + 4 \cdot 1 + 1 \cdot 2 = 15 \end{aligned}$$

Mean is 0 (all Brownian increments have zero mean).

(2) Compute $\int_0^T W_t \, \mathrm{d}W_t$.
Answer: $\int_0^T W_t \, \mathrm{d}W_t = \frac{1}{2}W_T^2 - \frac{1}{2}T$.

解析: Let $f(x) = \frac{1}{2}x^2$. By Ito's formula:

$$\begin{aligned} \mathrm{d}f(W_t) &= f'(W_t)\,\mathrm{d}W_t + \frac{1}{2}f''(W_t)(\mathrm{d}W_t)^2 \\ &= W_t \,\mathrm{d}W_t + \frac{1}{2}\,\mathrm{d}t \end{aligned}$$

So: $W_t \,\mathrm{d}W_t = \mathrm{d}\big(\frac{1}{2}W_t^2\big) - \frac{1}{2}\,\mathrm{d}t$

Integrate from 0 to T: $\int_0^T W_t \,\mathrm{d}W_t = \frac{1}{2}W_T^2 - \frac{1}{2}T$ (since $W_0 = 0$)

The correction term $-\frac{T}{2}$ comes from quadratic variation of Brownian motion.

(3) Assume $X_t = 3t^2 + W_t^4 + ctW_t^2$, prove X_t is a diffusion process.
Answer: $\mathrm{d}X_t = (4W_t^3 + 2ctW_t)\,\mathrm{d}W_t + \left[(6+c)t + (6+c)W_t^2\right]\mathrm{d}t$.

解析: Apply Ito's formula to each term:

- $\mathrm{d}\big(3t^2\big) = 6t\,\mathrm{d}t$
- $\mathrm{d}\big(W_t^4\big) = 4W_t^3\,\mathrm{d}W_t + 6W_t^2\,\mathrm{d}t$ (by Ito with $g(x) = x^4$, $g' = 4x^3$, $g'' = 12x^2$)

For the product term ctW_t^2, use the stochastic product rule $\mathrm{d}(U_tV_t) = U_t\,\mathrm{d}V_t + V_t\,\mathrm{d}U_t + \mathrm{d}U_t\,\mathrm{d}V_t$:

Let $U_t = t$, $V_t = W_t^2$.
- $\mathrm{d}U_t = \mathrm{d}t$
- $\mathrm{d}V_t = \mathrm{d}\big(W_t^2\big) = 2W_t\,\mathrm{d}W_t + \mathrm{d}t$ (by Ito with $g(x) = x^2$)

$$\begin{aligned} \mathrm{d}\big(tW_t^2\big) &= t\,\mathrm{d}\big(W_t^2\big) + W_t^2\,\mathrm{d}t + \mathrm{d}t\,\mathrm{d}\big(W_t^2\big) \\ &= t(2W_t\,\mathrm{d}W_t + \mathrm{d}t) + W_t^2\,\mathrm{d}t \\ &= 2tW_t\,\mathrm{d}W_t + \big(t + W_t^2\big)\,\mathrm{d}t \end{aligned}$$

So: $\mathrm{d}\big(ctW_t^2\big) = 2ctW_t\,\mathrm{d}W_t + c\big(t + W_t^2\big)\,\mathrm{d}t$

Combining all three components:

$$
\begin{aligned}
\mathrm{d}X_t &= 6t\,\mathrm{d}t + [4W_t^3\,\mathrm{d}W_t + 6W_t^2\,\mathrm{d}t] + [2ctW_t\,\mathrm{d}W_t + c(t+W_t^2)\,\mathrm{d}t] \\
&= (4W_t^3 + 2ctW_t)\,\mathrm{d}W_t + [6t + 6W_t^2 + ct + cW_t^2]\,\mathrm{d}t \\
&= (4W_t^3 + 2ctW_t)\,\mathrm{d}W_t + [(6+c)t + (6+c)W_t^2]\,\mathrm{d}t
\end{aligned}
$$

$$
\mathrm{d}X_t = \underbrace{(4W_t^3 + 2ctW_t)}_{\sigma(t,W_t)}\,\mathrm{d}W_t + \underbrace{[(6+c)t + (6+c)W_t^2]}_{\mu(t,W_t)}\,\mathrm{d}t
$$

This is in SDE form $\mathrm{d}X_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$, so X_t is an Ito process / diffusion process.

(4) If the drift of X_t is always 0, what is the possible value of c?
Answer: $c = -6$.

解析: From (3), the drift is: $\mu = (6+c)t + (6+c)W_t^2 = (6+c)(t+W_t^2)$.

For drift $\equiv 0$, we need $(6+c)(t+W_t^2) = 0$ for all t and all W_t.

Since $t + W_t^2$ is not identically zero, we must have $6 + c = 0$.

Therefore: $c = -6$.

When $c = -6$, the drift vanishes completely: $(6+(-6))(t+W_t^2) = 0(t+W_t^2) = 0$.
The diffusion process becomes a pure martingale: $\mathrm{d}X_t = (4W_t^3 - 12tW_t)\,\mathrm{d}W_t$.

(5) Is geometric Brownian motion a diffusion process? Why?
Answer: Yes. Geometric Brownian motion (GBM) is a special case of a diffusion process.

解析: The SDE form of GBM

$$
dS_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t
$$

fully conforms to the general definition of a diffusion process

$$
dX_t = \mu(t, X_t)\,\mathrm{d}t + \sigma(t, X_t)\,\mathrm{d}W_t
$$

where the drift coefficient $\mu(t,x) = \mu x$ and the diffusion coefficient $\sigma(t,x) = \sigma x$ are both continuous functions of t and x.

- The drift coefficient $\mu(t,x) = \mu x$ and diffusion coefficient $\sigma(t,x) = \sigma x$ satisfy the **linear growth condition** $|\mu(x,t)| + |\sigma(x,t)| \le C(1+|x|)$ and the **Lipschitz condition** $|\mu(x,t) - \mu(y,t)| + |\sigma(x,t) - \sigma(y,t)| \le D|x-y|$, guaranteeing existence and uniqueness of a strong solution to the SDE.
- By Itô's lemma, starting from the explicit solution $S_t = S_0 \exp\Big(\big(\mu - \frac{\sigma^2}{2}\big)t + \sigma W_t\Big)$, one can verify it satisfies the above SDE.
- The diffusion property of GBM guarantees the **Markov property** and **continuity** (continuous sample paths) of the asset price process, which is the theoretical foundation for pricing derivatives via PDEs (Feynman-Kac theorem) and the martingale pricing method.

课件参考: Chapter 2, Slides 46–47 (definition of diffusion process and SDE), Slide 49 (SDE of BSM and GBM), Slide 52 (examples of diffusion processes).

(6) How to convert the BSM to the risk-neutral measure?

Answer: Via a change of measure using the Girsanov theorem. Define the market price of risk $\theta = \frac{\mu - r}{\sigma}$ and construct a new Brownian motion $\mathrm{d}\widetilde{W}_t = \mathrm{d}W_t + \theta\,\mathrm{d}t$. Under the risk-neutral measure $\mathbb{Q}$, the BSM SDE becomes $dS_t = rS_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}\widetilde{W}_t$, and the discounted asset price $e^{-rt}S_t$ becomes a martingale.

解析: The BSM SDE under the real-world measure is

$$dS_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t$$

By the **Girsanov theorem**, take the constant market price of risk

$$\theta = \frac{\mu - r}{\sigma}$$

and define the Radon-Nikodym derivative process

$$Z(t) = \exp\left(\int_0^t \theta\,\mathrm{d}W_s - \frac{1}{2}\int_0^t \theta^2\,\mathrm{d}s\right), \quad \frac{d\mathbb{Q}}{d\mathbb{P}} = Z(t)$$

Then $\widetilde{W}_t = W_t + \theta t$ is a standard Brownian motion under $\mathbb{Q}$.

Substituting $\mathrm{d}W_t = \mathrm{d}\widetilde{W}_t - \theta\,\mathrm{d}t$ into the BSM SDE:

$$\begin{aligned} dS_t &= \mu S_t\,\mathrm{d}t + \sigma S_t\left(\mathrm{d}\widetilde{W}_t - \frac{\mu - r}{\sigma}\,\mathrm{d}t\right) \\ &= rS_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}\widetilde{W}_t \end{aligned}$$

The drift changes from μS_t to rS_t.

Economic intuition:

- Under $\mathbb{Q}$, the expected return of all assets equals the risk-free rate r.
- The discounted price process $e^{-rt}S_t$ is a martingale under $\mathbb{Q}$: $E^{\mathbb{Q}[e^{-rT}S_T \mid \mathcal{F}_t]} = e^{-rt}S_t$.
- Risk-neutral pricing formula: $V_t = e^{-r(T-t)}E^{\mathbb{Q}[V_T \mid \mathcal{F}_t]}$.
- The real-world expected return μ **completely disappears** from the pricing formula — consistent with the fact that the real probability p does not appear in the binomial tree pricing formula.

课件参考: Chapter 2, Slide 67 (Girsanov theorem), Slide 69 (change of measure for BSM and market price of risk), Slide 73 (martingale pricing principle of risk-neutral pricing).

(7) What is the biggest difference between the Itô integral and ordinary calculus?

Answer: The biggest difference is that the quadratic variation of Brownian motion is non-zero. In ordinary calculus $(\mathrm{d}x)^2$ is negligible, but in stochastic calculus $(\mathrm{d}W_t)^2 = \mathrm{d}t$, so the second-order term is of the same order as the first-order infinitesimal and cannot be neglected. This leads to an additional second-order correction term $\frac{1}{2}f''(X_t)\mathrm{d}[X,X]_t$ in Itô's lemma.

解析:

- **Quadratic variation of Brownian motion:** $[W,W]_t = t$, i.e. $(\mathrm{d}W_t)^2 = \mathrm{d}t$. This means $(\mathrm{d}W_t)^2$ is of order $\mathrm{d}t$, not $o(\mathrm{d}t)$.
- **Chain rule in ordinary calculus:** For a smooth function $x(t)$, $df(x(t)) = f'(x(t))\,\mathrm{d}x(t)$, because $(\mathrm{d}x)^2$ is a higher-order infinitesimal that can be discarded in the Taylor expansion.
- **Itô's lemma:** For a diffusion process X_t, since $\mathrm{d}[X,X]_t = \sigma^2(X_t,t)\,\mathrm{d}t$,

$$\mathrm{d}f(X_t) = f'(X_t)\,\mathrm{d}X_t + \frac{1}{2}f''(X_t)\mathrm{d}[X,X]_t$$

The second-order correction term $\frac{1}{2}f''(X_t)\mathrm{d}[X,X]_t$ is the core feature distinguishing Itô calculus from ordinary calculus.

Other key differences:

- **Left-endpoint evaluation:** The Itô integral $\int_0^T X_s\,\mathrm{d}W_s = \lim \sum X_{t_{j-1}}\left(W_{t_j} - W_{t_{j-1}}\right)$ evaluates the integrand at the left endpoint of each interval, ensuring non-anticipating (adapted) integrands. In the Riemann integral, the sampling point can be chosen arbitrarily.
- **Martingale property:** The Itô integral $\int_0^t X_s\,\mathrm{d}W_s$ is a martingale, which is crucial for risk-neutral pricing.
- **Itô isometry:** $E\left[\left(\int_0^t X_s\,\mathrm{d}W_s\right)^2\right] = E\left[\int_0^t X_s^2\,\mathrm{d}s\right]$, which has no counterpart in ordinary integration.
- **Pathwise meaning:** Brownian motion sample paths are nowhere differentiable and have unbounded variation, so the Itô integral cannot be defined pathwise and must rely on probabilistic limits in the L^2 space.

课件参考: Chapter 2, Slides 31–35 (non-differentiability, unbounded variation, bounded quadratic variation of Brownian motion), Slide 42 (left-endpoint definition of the Itô integral), Slide 44 (martingale property and Itô isometry), Slide 48 (Itô's lemma and the second-order correction term).

三、Risk-Neutral Pricing

(1) What is the Black-Scholes model?

Answer: The Black-Scholes (Black-Scholes-Merton) model is a mathematical framework for pricing European options based on no-arbitrage principles.

SDE under risk-neutral measure $\mathbb{Q}$: $\mathrm{d}S_t = rS_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}\widetilde{W}_t$.

Key assumptions: no arbitrage, continuous trading, constant volatility σ and risk-free rate r, no dividends, European exercise, log-normal prices, perfect markets.

BS PDE:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0$$

European Call formula: $C = S_0 N(d_1) - Ke^{-rT} N(d_2)$ **European Put formula:** $P = Ke^{-rT} N(-d_2) - S_0 N(-d_1)$

where: $d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}$ $d_2 = d_1 - \sigma\sqrt{T}$

解析: The model was developed by Fischer Black, Myron Scholes, and Robert Merton (1997 Nobel Prize). It assumes log-normal stock prices and complete markets, enabling replication and no-arbitrage pricing.

(2) Payoff function for ¥1 if $S_T > K$ at maturity T.

Answer: This is a cash-or-nothing binary call option.

$\Phi(S_T) = \mathbf{1}_{\{S_T > K\}} = \begin{cases} 1, \text{ if } S_T > K \\ 0, \text{ if } S_T \le K \end{cases}$

(3) Use Black-Scholes option pricing to price this option.

Answer: $V_0 = e^{-rT} N(d_2)$.

解析: Under $\mathbb{Q}$: $\mathrm{d}S_t = rS_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}\widetilde{W}_t$

Solving: $S_T = S_0 \exp\left(\left(r - \frac{\sigma^2}{2}\right)T + \sigma \widetilde{W}_T\right)$

Risk-neutral pricing: $V_0 = e^{-rT} E_{\mathbb{Q}[\mathbf{1}_{\{S_T > K\}}]} = e^{-rT} \mathbb{Q}(S_T > K)$

$$\begin{aligned} \mathbb{Q}(S_T > K) &= \mathbb{Q}\left(\ln\left(\frac{S_T}{S_0}\right) > \ln\left(\frac{K}{S_0}\right)\right) \\ &= \mathbb{Q}\left(\frac{\ln\left(\frac{S_0}{K}\right) + \left(r - \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}} > -Z\right) \quad \text{where } Z \sim N(0,1) \\ &= N\left(\frac{\ln\left(\frac{S_0}{K}\right) + \left(r - \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}\right) = N(d_2) \end{aligned}$$

Therefore: $V_0 = e^{-rT} N(d_2)$.

Each ¥1 payout is discounted at e^{-rT} and multiplied by the risk-neutral probability $N(d_2)$ that the option finishes in-the-money.

(4) If you hold a long American call option, what is your optimal exercise strategy?

Answer: Hold to maturity, never exercise early (assuming the underlying asset pays no dividends).

解析: Courseware theorem (Slide 79): **American call option and European call option (with the same parameters) have the same price, if there is no dividend.** That is, without dividends, the optimal exercise strategy for an American call is not to exercise (hold to maturity).

Reasons:

1. **Positive time value:** For any $t < T$, the market price of an American call strictly exceeds its intrinsic value: $C^{\mathrm{Am}}(S_t, t) > (S_t - K)^+$. Early exercise yields only the intrinsic value, forfeiting all time value.
2. **The option acts as "insurance":** Holding a call offers unlimited upside while limiting downside risk to the premium; early exercise exposes the holder directly to the underlying's price risk.
3. **Interest benefit of delayed payment:** Early exercise pays K at time t, while holding to maturity pays K at time T. Delaying payment earns risk-free interest $K\left(1 - e^{-r(T-t)}\right) > 0$.
4. **Price lower bound inequality:** $C^{\mathrm{Am}}(S_t, t) \geq C^{\mathrm{Eur}}(S_t, t) \geq S_t - Ke^{-r(T-t)} > S_t - K$ holds for all $t < T$. Selling the option in the market is always superior to exercising.

Exception: When the underlying pays dividends, early exercise immediately before the ex-dividend date may be optimal (exercising captures the dividend but forfeits remaining time value). At all other times, holding remains optimal.

Contrast: For an American put, even without dividends, early exercise may be optimal (when S_t is sufficiently low, the interest on K may exceed the time value), requiring optimal stopping theory to determine the optimal exercise boundary.

课件参考: Chapter 2, Slide 79 (equivalence theorem of American and European calls), Slide 57 (intrinsic value and time value), Slide 76 (Put-Call Parity), Slides 78–80 (optimal stopping and American put exercise boundary).